

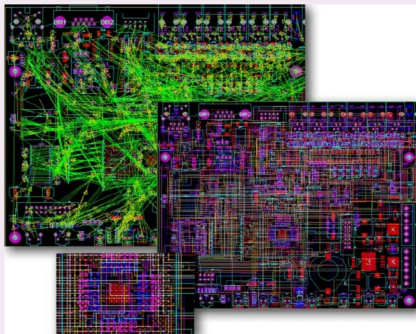


Fig. 2: Shape based auto router before and after routing on a six-layer PCB

interrupts, timers, universal synchronous/asynchronous receiver/transmitters and all other peripherals present on each supported processor.

Giving you the flexibility to interact with your design using onscreen indicators such as LCD and LED displays, and actuators, you can apply your firmware to the microcontroller of your choice and co-simulate the program within a mixed-mode simulation program with integrated circuit emphasis (SPICE) circuit simulation. If coding is your forte, take the design flowchart back to its Arduino C++ code, or convert it to an Arduino sketch and play with the code.

A resourceful kit. Proteus design kit contains thousands of common parts you would need. It allows you to import more using BSD format. The physical and electrical designs can be viewed as grids, paving the way for easier cross-probing and component or net selection. Hierarchical designs and system level designs are just a few clicks away.

A powerful PCB design suite

Combining the schematic capture and ARES PCB layout programs, this

design suite targets PCB designs with a professional touch. The next section discusses a few of the standard features of this layout designer.

In addition to these, the tool supports 16 copper layers, 10nm resolution and any angle placement. There is also support for power planes, letting you place polygonal regions within which inner boundaries are automatically created around existing pads and tracks.

Intelligent routing. The wiring process is intuitive. Called Follow-Me Routing, the control is in your hands, giving you the best-fit route within designated constraints. As you move the mouse, so does the wiring, linking together the parts of your schematic. And once a base is done, you can create a design snippet to import into a future project.

The snippet can range from a simple rectifier circuit to the modeling of a USB engine, or can simply be a template containing basic layout information like boundaries and holes. It goes without saying that schematic hierarchies and logic schematic designs are fully supported by the tool.

Go the manual way... The one thing layout designers have a tough time getting right is congestion release—tight spaces with too many wires that can neither be re-routed easily nor done without. It is a matter of trial and error, a process that is eased a bit in Proteus by letting you add via connections to pads, aka surface mount technology.

Manual routing can also create curved tracks by holding down ctrl key, and thick tracks are automatically necked to fit in with the set rules. You can also logically or globally control the size and diameter of tear-drops, as required. To make sure you do not violate rules, a live status for connectivity and design rules can be found at the bottom of the screen.

... Or exploit the auto feature.

What the auto-router gives you is flexibility. How is that? Because auto-routing is shape based. A quick and an easy fix to your design rule violations! Using advanced cost based conflict-reduction algorithms proven to maximise completion rates on even the most densely-packed boards, the router gives you the best routing possible under given circumstances. Advanced-level users also drive the router either by writing custom routing scripts or by directly entering routing commands interactively, to route particular areas or specify fan-out criteria.

Handling large numbers of pins

Schematic Capture is designed to aid a high degree of control over the drawing appearance, be it line widths, fill styles or fonts; extending this to the layout graphics, too. Dealing with a ball-grid array or a high pin count device can be tricky. The sheer volume complicates everything from connection to routing.

To handle this easily, Proteus lets you first bring in the component definition directly from the manufacturer's boundary scan description language file. You can then split these into smaller, more manageable elements